Artificial Intelligence (AI)

LAB 2 (Part 1)

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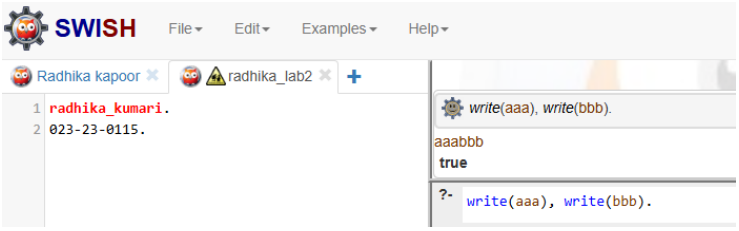
CMS: 023-23-0115

Task 1: Predict the output.

- `write(aaa), write(bbb).`

  aaabbb

  true.

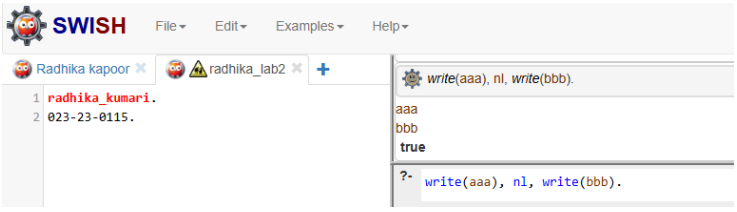


- `write(aaa),nl, write(bbb).`

  aaa

  bbb

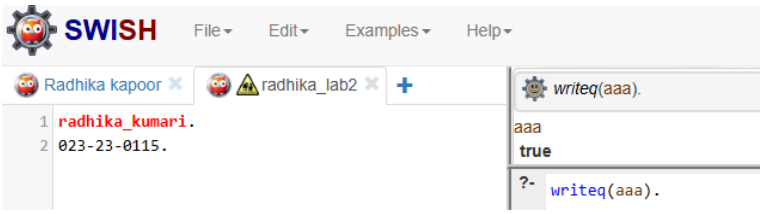
  true.



- `writeq(aaa).`

  aaa

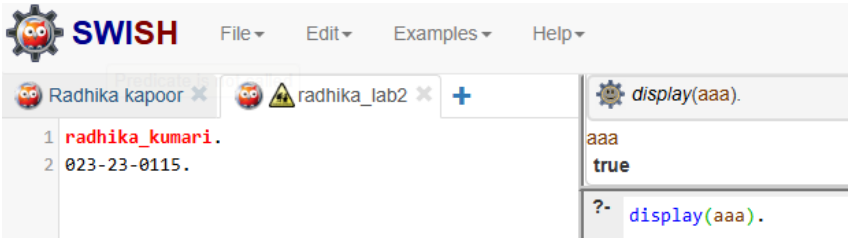
  true.



- display(aaa).

  aaa

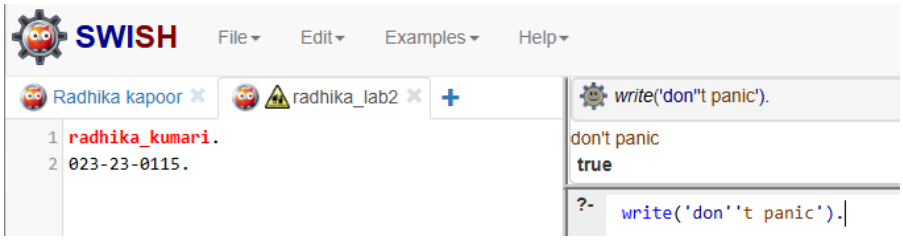
  true



- write ('don''t panic').

  don't panic

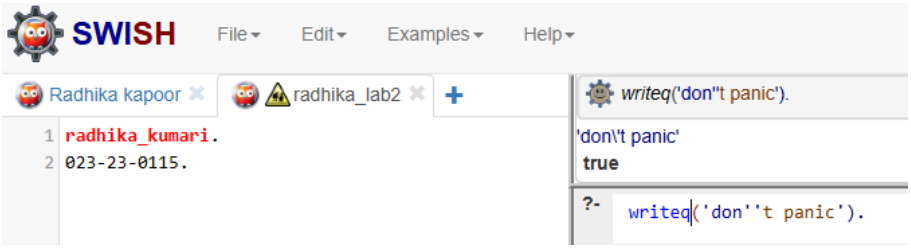
  true



- writeq ('don''t panic').

  'don\'t panic'

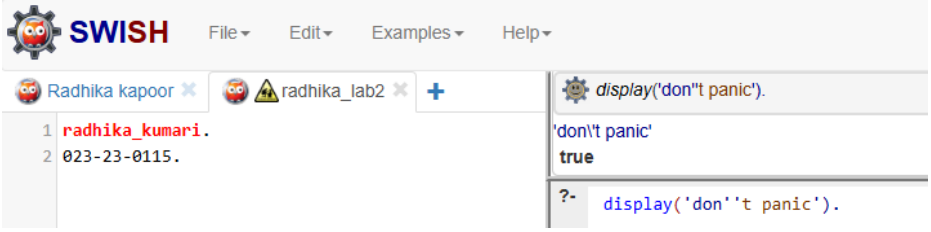
  true



- display ('don''t panic').

  don't panic

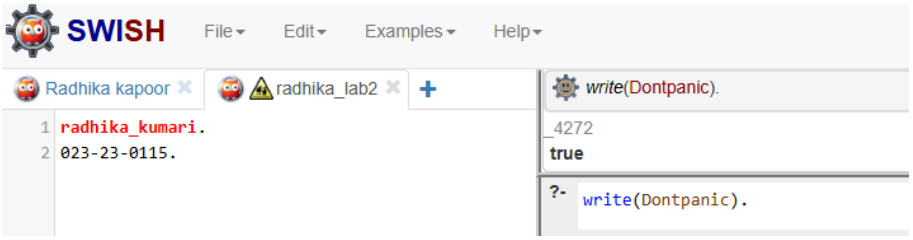
  true



- write (Dontpanic).

  _4272

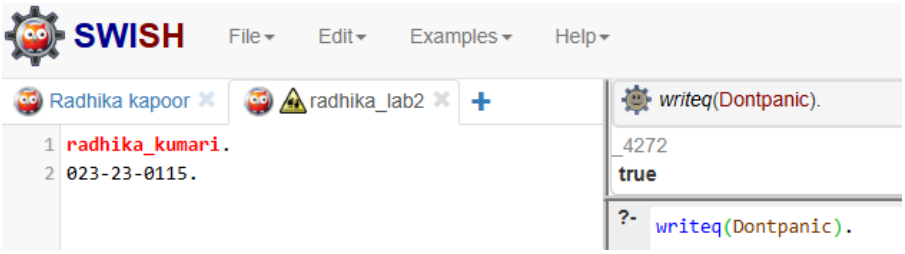
  true



- writeq (Dontpanic).

  _4272

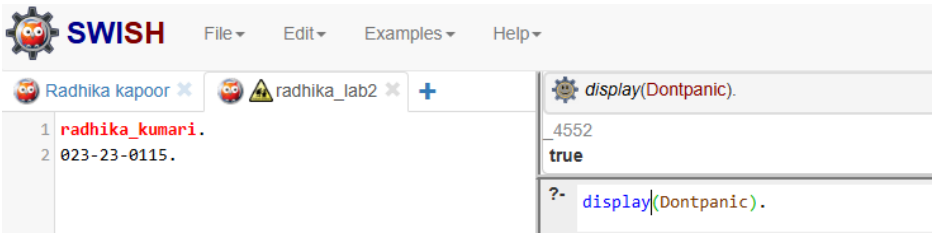
  true



- display (Dontpanic).

  _4552

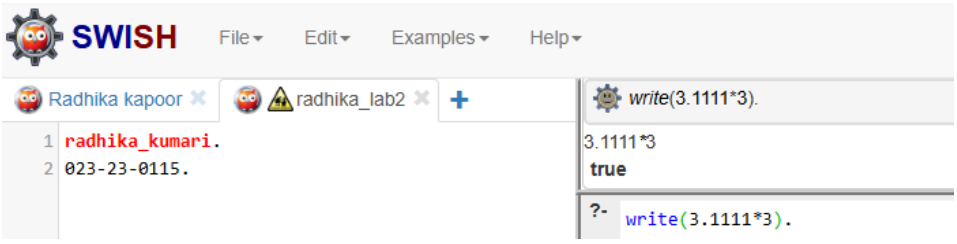
  true



- write (3.1111*3).

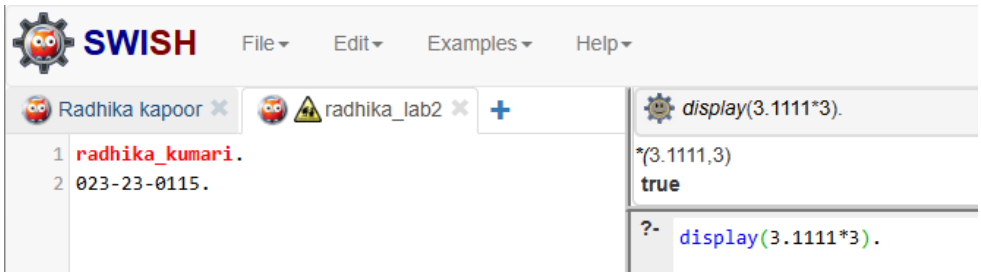
  3.1111*3

  true



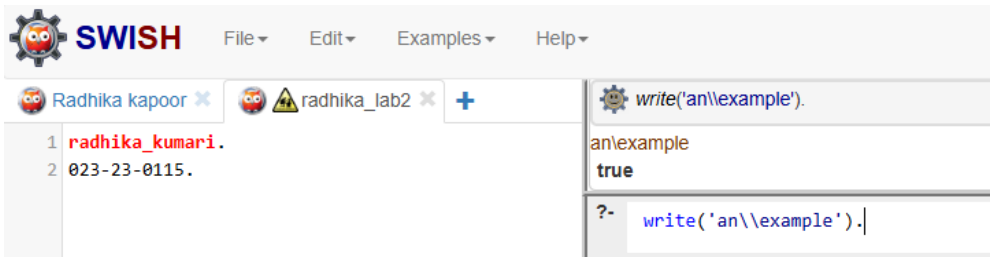
- display (3.1111*3).

```
*(3.1111,3)

true
```



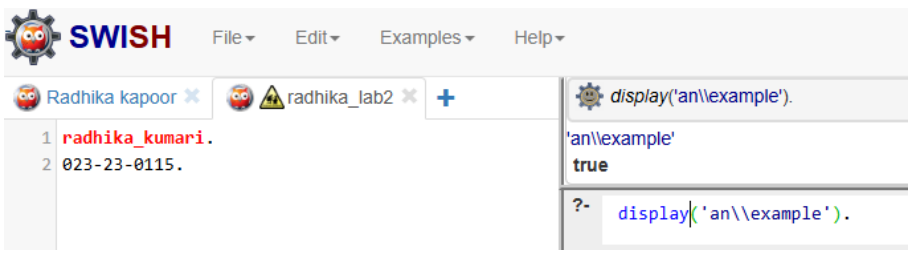
- write('an\\example').

```
an\example

true
```



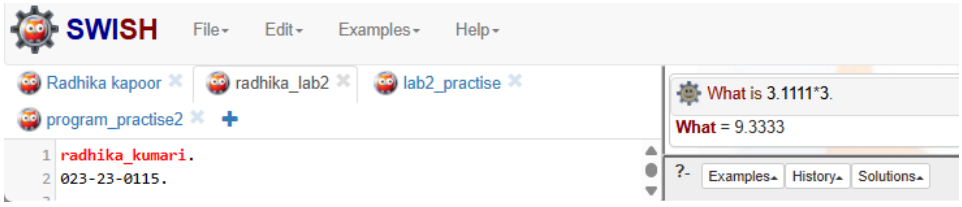
- display('an\\example').

```
'an\\example'

true
```



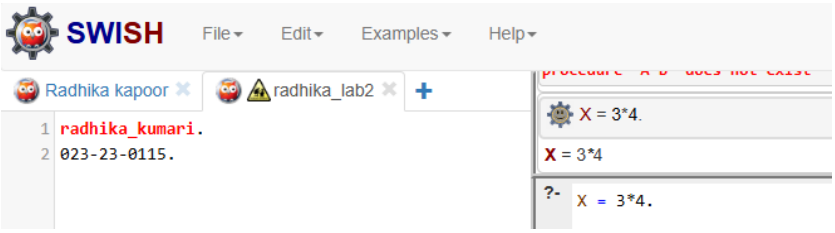
- What is 3.111*3.

```
What = 9.3333
```

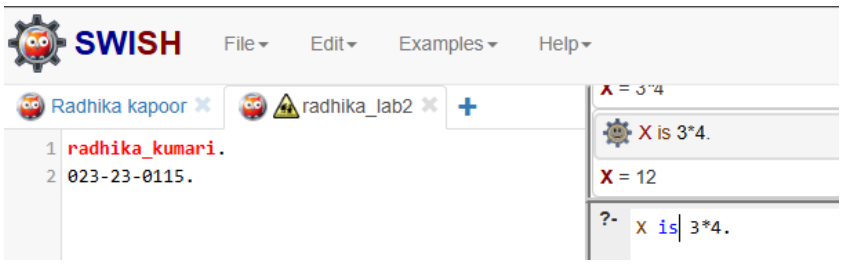


Task 2: Prolog respond to the following queries.

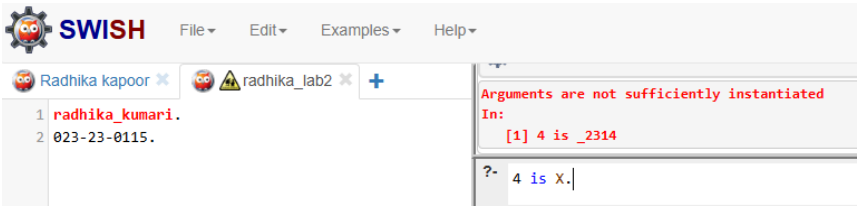
1. X = 3*4.



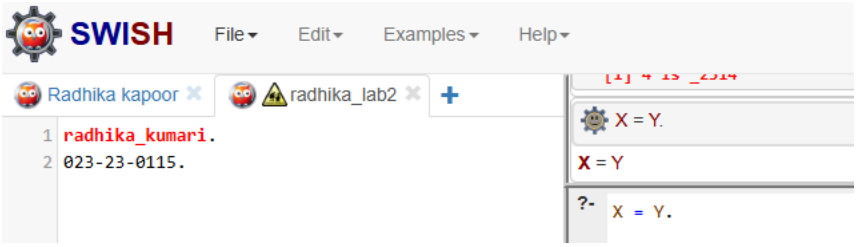
2. X is 3*4.



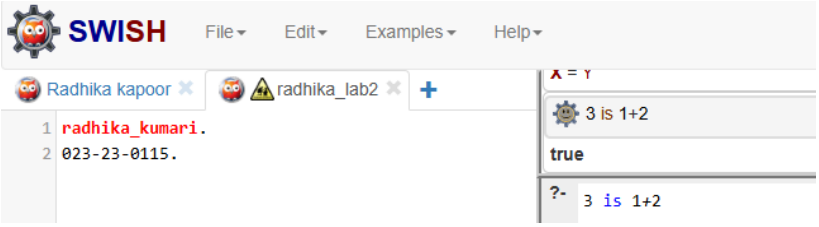
3. 4 is X.



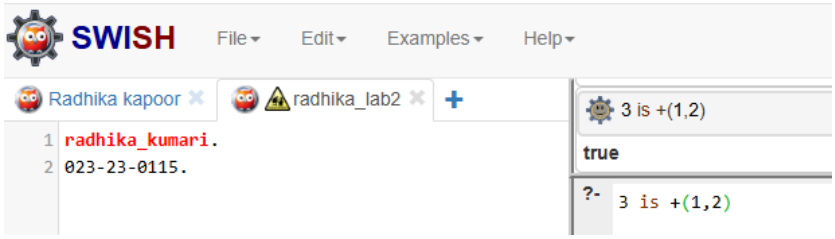
4. X = Y.



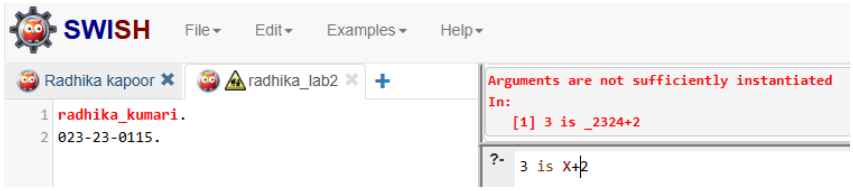
5. 3 is 1+2.



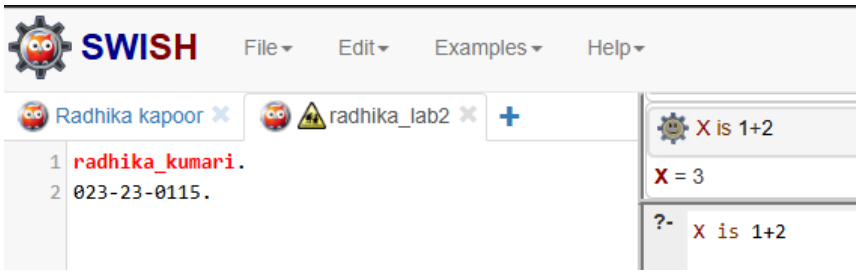
6. 3 is +(1,2).



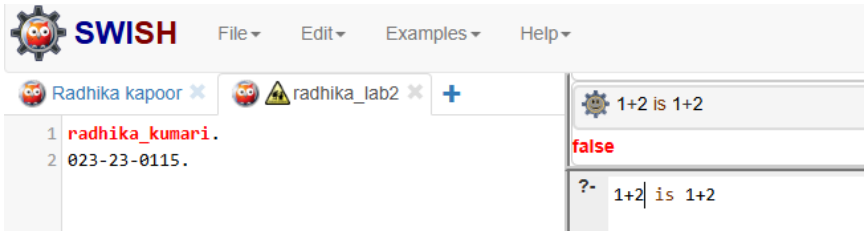
7. 3 is X+2.



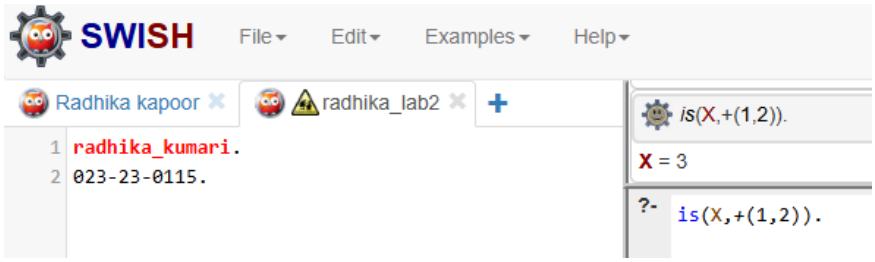
8. X is 1+2.



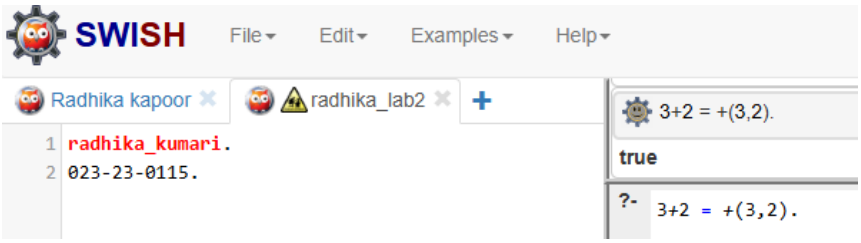
9. 1+2 is 1+2.



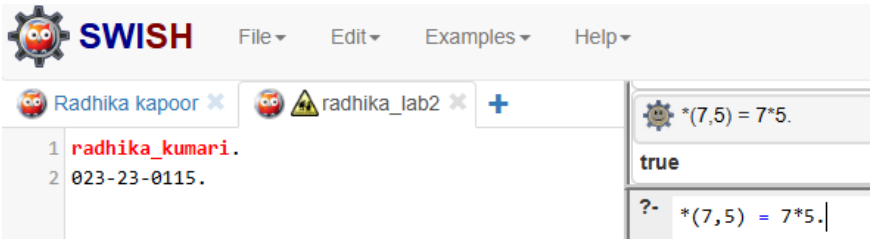
10. is (X,+(1,2)).



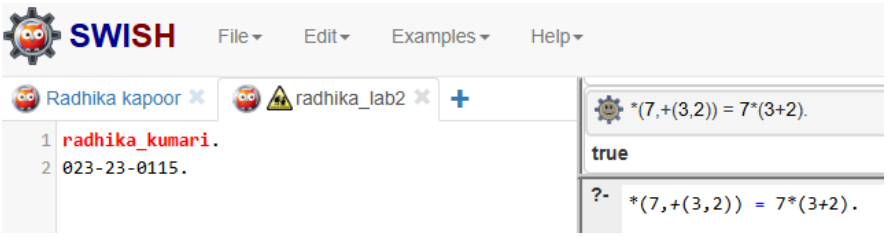
11. 3+2 = +(3,2).



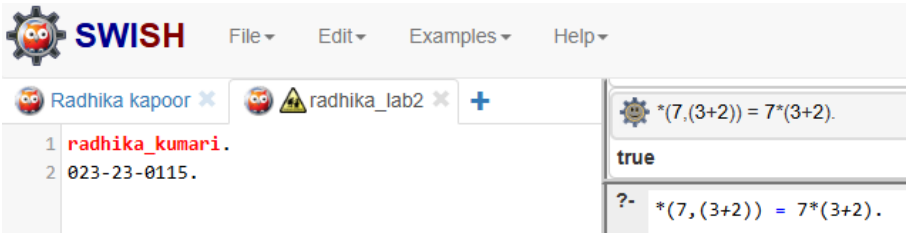
12. *(7,5) = 7*5.



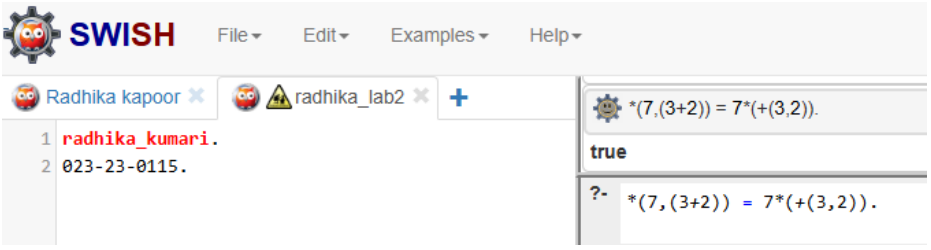
13. *(7,+(3,2)) = 7*(3+2).



14. *(7,(3+2)) = 7*(3+2).



15. *(7,(3+2)) = 7*(+(3,2)).



LAB 2 (Part 2)

- **Write a Prolog database containing data on the club's members by using rules to facilitate retrieval of the following information:**

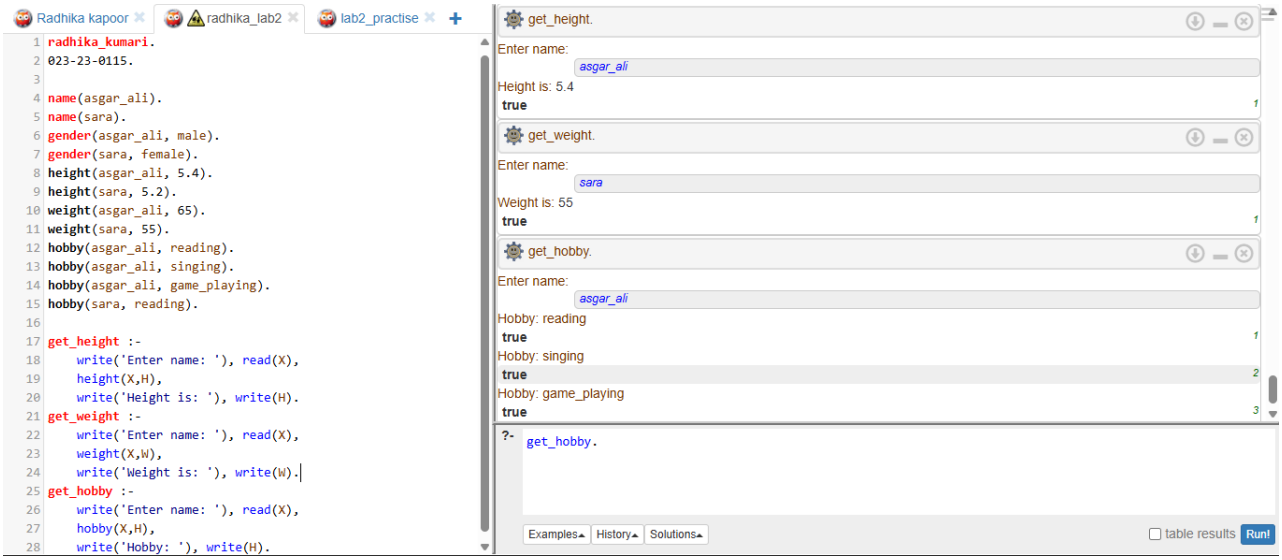
  a. Given a name, find height or weight or hobbies.

Facts:

```
name(asgar_ali).
name(sara).
gender(asgar_ali, male).
gender(sara, female).
height(asgar_ali, 5.4).
height(sara, 5.2).
weight(asgar_ali, 65).
weight(sara, 55).
hobby(asgar_ali, reading).
hobby(asgar_ali, singing).
hobby(asgar_ali, game_playing).
hobby(sara, reading).
```

Rules:

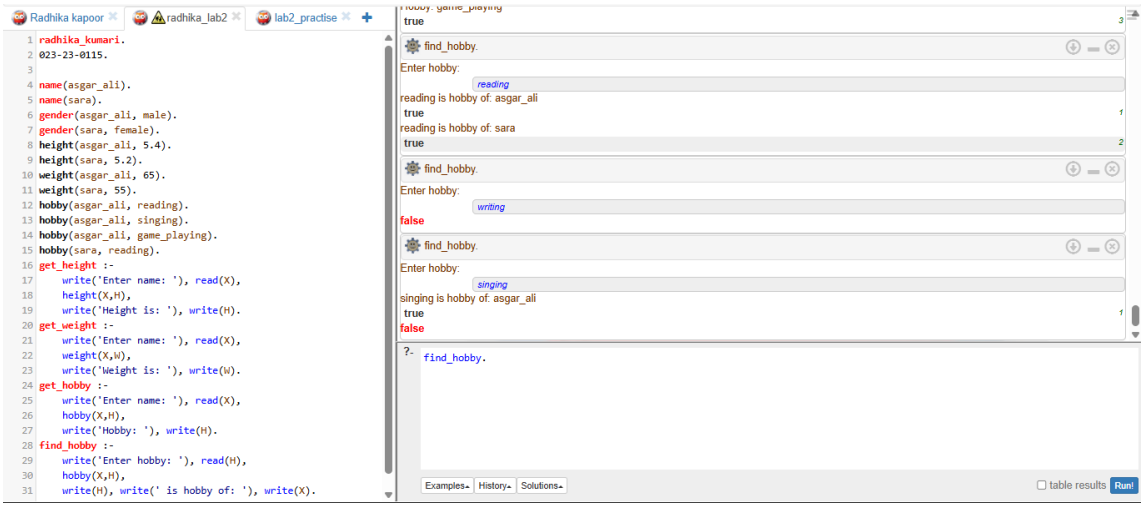
```
get_height :-
write('Enter name: '), read(X),
height(X,H),
write('Height is: '), write(H).
get_weight :-
write('Enter name: '), read(X),
weight(X,W),
write('Weight is: '), write(W).
get_hobby :-
write('Enter name: '), read(X),
hobby(X,H),
write('Hobby: '), write(H).
```



b. Find someone with a particular hobby.

Rules:

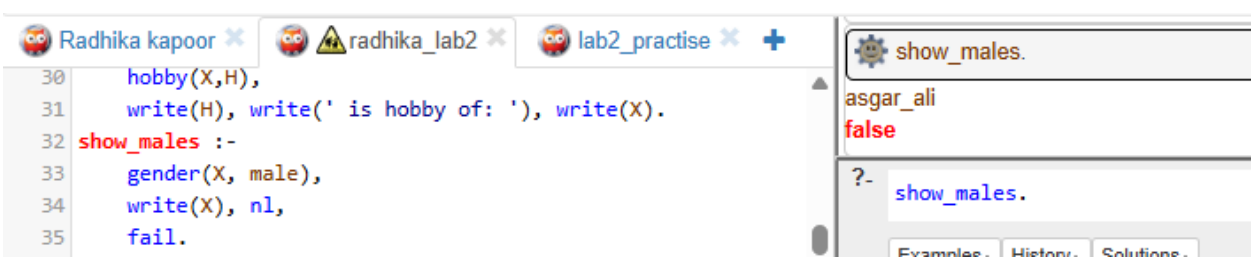
```
find_hobby :-
    write('Enter hobby: '), read(H),
    hobby(X,H),
    write(H), write(' is hobby of: '), write(X).
```



c. Show all the male members from facts by using backtracking method.

Rules:

```
show_males :-
    gender(X, male),
    write(X), nl,
    fail.
```



- **<u>Write a prolog program that will get value from user and show it's previous and next value in sequence as in given output.</u>**

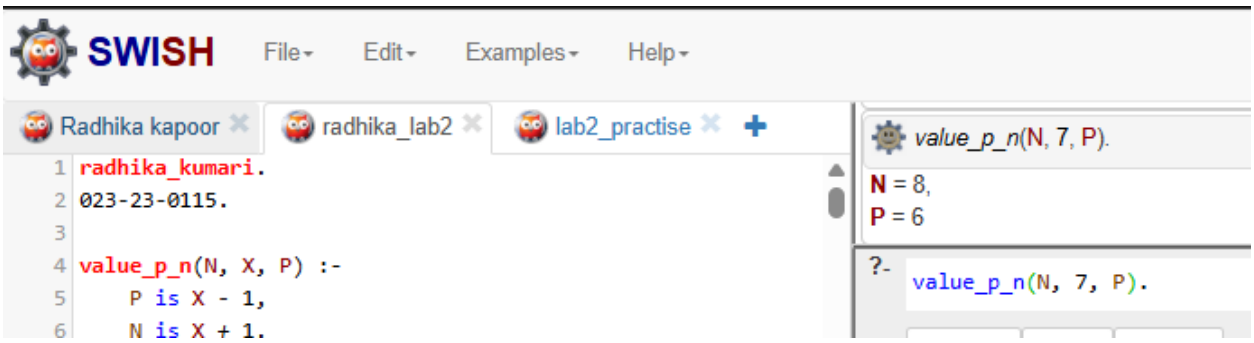
a. <u>With constant given value.</u>

<u>Rules:</u>

```
value_p_n(N, X, P) :-
    P is X - 1,
    N is X + 1.
```

<u>Query:</u>

```
value_p_n(N, 7, p).
```



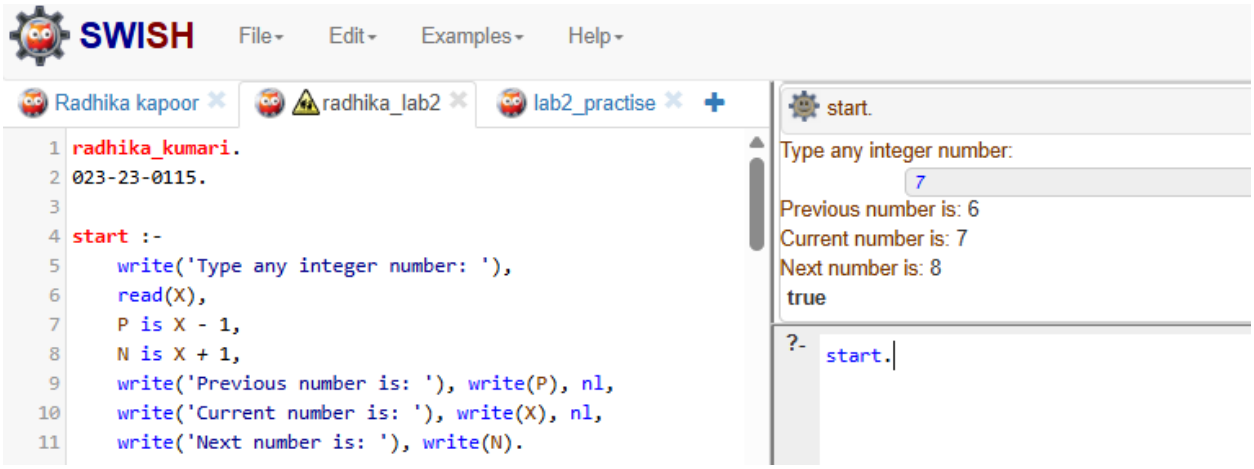
b. <u>Through getting inputs.</u>

<u>Rules:</u>

```
start :-
    write('Type any integer number: '),
    read(X),
    P is X - 1,
    N is X + 1,
    write('Previous number is: '), write(P), nl,
    write('Current number is: '), write(X), nl,
    write('Next number is: '), write(N).
```

<u>Query:</u>

```
start.
```



Rules Exercise:

Exercise 1:

Indicate whether the following are syntactically correct Rules. Write answer Yes or No.

- a :- b, c, d:- e f.

  Is this syntactically, correct? **NO**

- happy(X):- a , b.

  Is this syntactically, correct? **YES**

- happy(X):- hasmoney(X) & has friends(X).

  Is this syntactically, correct? **NO**

- fun(fish):- blue(betty), bike(yamaha).

  Is this syntactically, correct? **YES**

Exercise 2:

Given the database below, study the queries underneath it. Indicate whether you think a

particular query will succeed or fail by answer yes or no.

Facts:

```
likes(john,mary).
likes(john,trains).
likes(peter,fast_cars).
hobby(john,trainspotting).
hobby(tim,sailing).
hobby(helen,trainspotting).
hobby(simon,sailing).
```

Rule:

```
likes(Person1,Person2):-
      hobby(Person1,Hobby),
      hobby(Person2,Hobby).
```

- ?- likes(john,trains). Will this Succeed? **YES**
- ?- likes(helen,john). Will this Succeed? **YES**
- ?- likes(tim,helen). Will this Succeed? **NO**
- ?- likes(john,helen). Will this Succeed? **YES**

Exercise 3:

How many facts, rules, clauses, and predicates are there in the following knowledge base? What are the heads of the rules, and what are the goals they contain?

Facts: **3**

```
woman(nimmy).
woman(maria).
man(jabbaar).
```

Rules: **4**

```
person(X) :- man(X); woman(X).
loves(X,Y) :- knows(Y,X).
father(Y,Z) :- man(Y), son(Z,Y).
father(Y,Z) :- man(Y), daughter(Z,Y).
```

Clauses: **7**

Exercise 4:

Represent the following in Prolog:

- Bashir is a killer.

  ```
  killer(bashir).
  ```

- Maria and Maqbool are married.

  ```
  married(maria, maqbool).
  ```

- Zaib is dead.

  ```
  dead(zaib).
  ```

- Maqbool kills everyone who gives Maria a foot massage.

  ```
  kills(maqbool, X) :-
        foot_massage(X, maria).
  ```

- Maria loves everyone who is a good dancer.

  ```
  loves(maria, X) :-
        good_dancer(X).
  ```

- Jabbaar eats anything that is nutritious or tasty.

  ```
  eats(jabbaar, X) :-
        nutritious(X); tasty(X).
  ```

Exercise 5:

Suppose we are working with the following knowledge base:

Facts:

```
wizard(ronny).
hasStick(harry).
footballPlayer(harry).
```

Rules:

```
wizard(X) :-
     hasBroom(X), hasStick(X).
hasBroom(X) :-
     footbalPlayer(X).
```

How does Prolog respond to the following queries?

- `wizard(ronny).` **Yes**
- `witch(ronny).` **NO**
- `wizard(amar).` **NO**
- `witch(amar).` **NO**
- `wizard(harry).` **YES**
- `wizard(Y).` **Y = roony; Y = harry**
- `witch(Y).` **NO**